SentinelLab

Experimental SOC / SIEM Investigation Workspace

Document purpose: public-facing technical overview and practical orientation; not a security certification or independent audit.

Project status: Experimental alpha; production validation incomplete

Technology / architecture: React, Python, FastAPI, MongoDB, Tauri 2, Docker, GitHub Actions

Architecture & capabilities

• Event ingestion: JSON, JSONL, CSV, Syslog and CEF

• Normalized event explorer, detection rules, alerts and case investigations

• Threat hunting, IOC workflows, MITRE ATT&CK; mapping and reports

• Role-aware access, workspace boundaries and audit trail

• Optional local or compatible AI and outgoing webhook configuration

Evaluation / getting started

• Use a local test environment and synthetic or authorized event records

• For desktop evaluation install MongoDB separately, bound to loopback

• Import sample events, review alerts, preserve exported evidence and record timestamps

Limitations and security considerations

• Not a production-certified SIEM

• CI packaging on Linux and macOS is not runtime acceptance

• Some threat-intelligence connectors are placeholders; response playbooks do not perform live EDR containment

Engineering roadmap

• Complete reproducible SOC case studies

• Improve event detail resizing and desktop acceptance

• Broaden detection regression and integration validation

Source and verification

Repository: https://github.com/ibrahim1101/SentinelLab

Information reflects the accessible project README and the supplied engineering resume. Features may change; verify current repository branches, releases and documentation before deployment. Do not upload private keys, user data, internal logs or unredacted evidence.